The Rank the Task Demands: A Causal Rank Law for Matrix Memories Trained on Group Composition

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Abstract

Matrix-valued memories make rank the natural budget of a learned representation: the number of independent directions a state spans bounds what it can bind, compose, and track. We report causal evidence, from two synthetic testbeds trained under a hard single-state bottleneck with an exact-continuous-recovery readout, that gradient descent recruits precisely the rank the task's algebra demands. On K-pair associative binding, where exact recovery provably requires state rank at least K, trained states develop effective rank tracking K (8.20 at K=8, 15.08 at K=16 for d=16), and train-time rank caps confirm the bound causally: at d=8, K=4, rank 3 yields zero exact recovery while rank 4 yields 0.94. On group-composition state tracking over five groups spanning the solvable/non-solvable divide, the recruited rank equals the group's minimal faithful real representation dimension $d_{\min}$ (Spearman $\rho = 0.9747$, the design's tie-capped maximum), the dimension-matched solvable/non-solvable pair S_4/A_5 is statistically equivalent under a pre-registered equivalence test, and a force-rank razor lands a step function at $d_{\min}$ in all five groups: one rank below, exact recovery is 0.000 everywhere, including every seed of a four-seed extension, while at $d_{\min}$ recovery returns to unconstrained-anchor levels. Representation theory, not computational complexity class, sets what gradient descent buys.

Keywords: effective rank, fast weights, group representations, state tracking

1. Introduction

A matrix-valued memory represents by spanning: whatever a $d \times d$ state binds or tracks, it holds in the geometry of its column space, and rank is the budget it spends. Prior work treats this budget descriptively (Nazari and Rusch, 2026; Sun et al., 2026) or through hand-built constructions (Nichani et al., 2025); neither answers what a geometric account of learned representations needs answered: when a task's algebra fixes a minimal representational dimension, does gradient descent recruit exactly that dimension, and is it causally load-bearing?

We answer both halves affirmatively under one discipline: a hard single-state bottleneck (the decoder reads only one matrix Z, verified by a gradient blank-out test) and exact-continuous-recovery scoring (cosine against the true continuous target, never argmax over a codebook, which would let a rank-1 state recover on the order of d associations (Nichani et al., 2025)). Section 2 includes the provable K-pair-binding foundation. The headline is group-composition state tracking, where representation theory supplies the ground truth: across five finite groups spanning the solvable/non-solvable divide, the minimal faithful real representation dimension $d_{\min}$ predicts the recruited rank almost perfectly, a matched-dimension solvable/non-solvable pair lands statistically equivalent (Section 3), and a pre-registered force-rank razor finds recovery exactly 0.000 one rank below $d_{\min}$ in every group, returning at $d_{\min}$ (Section 4): $d_{\min}-1$ is fatal, $d_{\min}$ suffices. An earlier razor sweep nulled through an instrument defect, an ambient identity block taxing every rank-capped arm; the raw-artifact diagnosis, the registered inconclusive verdict, and the repaired wave that landed the pre-registered prediction are part of the result (Appendix C).

2. Tasks, Models, and the Rank Instrument

Tasks. In the binding task, each episode presents K key–value pairs, the encoder writes one state $Z \in \mathbb{R}^{d\times d}$, prediction is the literal unbind Zk_j, and rec@0.9 is the fraction of queries with cosine to the true value above 0.9. The group task is the word problem of a finite group G: a word $w = g_{i_1} \cdots g_{i_L}$, drawn as a random walk on G's Cayley graph ($L \sim \mathrm{U}\{1,\dots,8\}$), must map to $\rho_G(g_{i_1} \cdots g_{i_L})$ under a pinned orthogonal reference representation of dimension $d_{\min}(G)$, the minimal faithful real representation dimension. The groups are S_3, S_4, A_5, S_5, A_6 with $d_{\min} = 2, 3, 3, 4, 5$ (Appendix A); the first two are solvable, the rest are not, and non-solvable word problems are NC^1-complete (Barrington, 1989). $d_{\min}$ is a different, representation-theoretic axis; S_4/A_5, matched at $d_{\min} = 3$ with opposite solvability, are the designed dissociation pair.

Model. A Transformer encoder with d_{state} learned row-reader queries maps each word to a single state $Z(w) \in \mathbb{R}^{d_{\text{state}}\times d_{\text{state}}}$, $d_{\text{state}} = d_{\min} + 2$, leaving spare dimensions so over-recruitment is expressible. The target embeds the reference block-diagonally ($\rho_G(\cdot) \oplus I_2$ in the first sweep, $\rho_G(\cdot) \oplus 0$ in the repaired wave); the loss is cosine distance, the readout is fixed with no learned weights that could launder rank, and the decoder reads only Z (blank-out verified). Per-group step budgets (8k–40k) were pinned by pre-registered convergence bars.

Instrument. The learned state is defined only up to scale and an orthogonal change of basis (a Schur intertwiner): $Z(w) \approx c\,Q\,[\rho_G(w) \oplus \cdot\,]\,Q^\top$. We estimate the model's own dominant $d_{\min}$-subspace U from the SVD of the *centered* covariance of $Z(w)$ over held-out words, measure **restricted effective rank** (entropy effective rank of $U^\top Z(w)U$), and score **degauged recovery**: cosine after fitting (c, Q) on a fitting split, evaluated on a disjoint split, reported as rec@0.9. Centering is load-bearing (Appendix B); rank is invariant under the fitted gauge, so a rank-deficient state cannot be degauged into a full-rank one. For force-rank cells the decisional readout was pinned, before the razor wave ran, to the conservative

full-Q Procrustes variant (*crosscheck* rec@0.9).

The provable foundation (binding).

Fact 1 *If $Zk_j = v_j$ exactly for K bindings with linearly independent keys and values, then*

$ZK_{\mathrm{mat}} = V_{\mathrm{mat}}$, so $K = \mathrm{rank}(V_{\mathrm{mat}}) \leq \mathrm{rank}(Z)$.

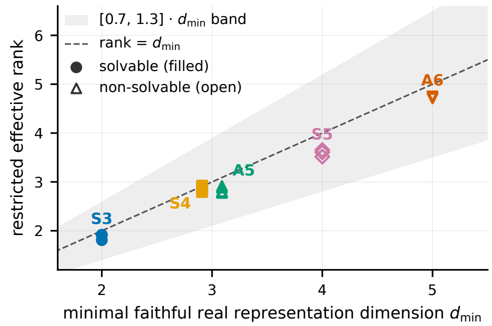
The bound is classical (Kohonen, 1972; Anderson, 1972) and holds only under exact continuous recovery. Gradient descent recruits the rank: at $d = 16$ the trained state's effective rank reaches 8.20 at $K = 8$ and 15.08 at $K = 16$, and the recruited rank is causally necessary: a spectral rank cap at $d = 8$, $K = 4$ leaves exact recovery at or below 0.0004 for every cap $k \leq 3$ and restores it to 0.940 at $k = 4$, a step exactly at the provable bound; the same operator composes exactly (Z^h, held-out depths to 21) in four of five seeds.

3. The Rank Law on Group Composition, Observed

Figure 1 (left) gives the observational leg: per-group mean restricted effective rank lands within 4–8% of d_{min} at every group, and all 19 seeds sit inside the pre-registered $[0.7, 1.3] \cdot d_{\mathrm{min}}$ band (means and per-seed values in Appendix B).

Spearman correlation between recruited rank and d_{min} is $\rho = 0.9747$, the maximum this design can express, since the S_4/A_5 tie caps ρ below 1; under the exact permutation null, $P(\rho \geq 0.8) = 8/120 \approx 6.7\%$, so by pre-registration this leg is corroborating rather than independently decisive (a disclosed length-robustness split is in Appendix B).

Dimension, not solvability. If recruited rank were set by computational class, the marquee pair should separate; if by representation dimension, coincide. A pre-registered Welch two-one-sided equivalence test on restricted effective rank (margin ± 0.5 rank-units, half the d_{min} ladder spacing; $n = 5$ per group; simulated power 100% to reject at a 1.0



G	anchor	$k{=}d_{\min}{-}1$	$k{=}d_{\min}$	$k{=}d_{\min}{+}1$	bar
S_3	0.550	0.000	0.450*	0.550	0.495
S_4	0.650	0.000	**0.800**	0.950	0.585
A_5	0.700	0.000	**0.700**	0.750	0.630
S_5	0.500	0.000	**0.600**	0.550	0.450
A_6	0.650	0.000	**0.650**	0.700	0.585

Figure 1: Left: recruited rank follows representation dimension, not solvability. Each point is one seed's restricted effective rank vs. its group's $d_{\min}$ (filled: solvable; open: non-solvable) with the pre-registered $[0.7, 1.3] \cdot d_{\min}$ band and the identity line; all 19 seeds are in band, and S_4/A_5 coincide at $d_{\min} = 3$. Right: the causal razor on the repaired rank-$d_{\min}$ target (crosscheck rec@0.9 by force-rank k, one seed per cell, unconstrained anchor, pre-registered 0.9×anchor bar): exactly 0.000 one rank below $d_{\min}$ everywhere, anchor-class at $d_{\min}$ (bold: clears the bar). *extended to four seeds under the pre-stated ±0.05 trigger: seed-mean 0.5625 clears the fixed 0.495 bar.

rank-unit gap) observes a difference of +0.019 rank-units (se 0.037, df 7.8), with both one-sided tests passing at roughly seven times the critical value ($t = 13.06$ and 14.12 against $t_{\text{crit}} = 1.865$): equivalence is declared decisively. The pair lands together on dimension, where the rank law puts it.

4. The Causal Razor

The decisive test intervenes on rank: the encoder's output is spectrally truncated to rank $k \in \{d_{\min}{-}1, d_{\min}, d_{\min}{+}1\}$ throughout training, against the zero-padded target $\rho_G(\cdot) \oplus 0$ of rank exactly $d_{\min}$, alongside an unconstrained anchor from the same family. Pre-registered reading: if $d_{\min}$ is load-bearing, $k = d_{\min}{-}1$ must fail and $k \geq d_{\min}$ must recover past 0.9× the anchor's crosscheck rec@0.9; a single below-$d_{\min}$ cell recovering falsifies necessity.

Figure 1 (right) and Figure 2 show the result on the pre-pinned crosscheck readout. Necessity is exact and noiseless: $k = d_{\min}{-}1$ yields rec@0.9 = 0.000 in all five groups, and in all four independent seeds of the S_3 extension, while the same cells' recovery cosines (0.61–0.84) show the zero is a threshold phenomenon, not a training collapse. Sufficiency holds at $d_{\min}$: S_4, A_5, S_5, A_6 clear their bars outright, and S_3, whose decisive cell landed inside the pre-stated ±0.05 marginality trigger, was extended to four seeds under the trigger's pre-registered routing: seed-mean 0.5625 against the fixed 0.495 bar (the literal fixed before the extension ran, avoiding a self-referential bar) confirms the fifth group. Both marquee groups confirm; the pre-registered causal criterion is met.

An earlier 58-cell sweep of this grid nulled through an instrument defect: its eye-padded target ($\rho_G(\cdot) \oplus I_2$, rank d_{state}) taxed every capped arm, whose 39 cells sat at their rank-k

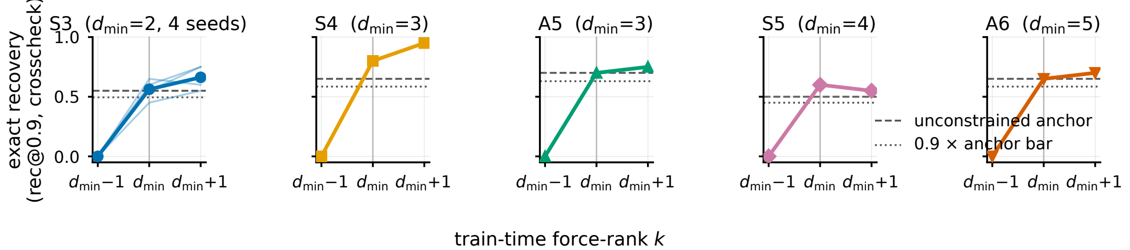


Figure 2: Exact recovery is a step function at $d_{\min}$: per group, crosscheck rec@0.9 on held-out words at force-rank $k \in \{d_{\min}-1, d_{\min}, d_{\min}+1\}$ (one seed per cell), vs. the unconstrained anchor (dashed) and the 0.9×anchor bar (dotted). S_3 overlays its four seeds (thin lines; bold mean); its below-$d_{\min}$ zero is unanimous.

optimum $\sqrt{k/d_{\text{state}}}$ (mean deviation 0.028); that verdict was registered as inconclusive with the tax as mechanism, and the zero-padded wave above is the registered fix (Appendix C).

5. Related Work and Limitations

Nazari and Rusch (2026) and Sun et al. (2026) measure effective-rank dynamics in pretrained linear-attention models, observationally, with no task algebra fixing a required dimension; Mishra et al. (2026) train matrix-state RNNs on S_3 composition but measure no state rank and run no rank intervention. Grazzi et al. (2025), Siems et al. (2025), and Merrill et al. (2024) characterize which word problems recurrent architectures can express, the solvability axis; the marquee equivalence shows the *learned geometry* sorts by representation dimension, not that axis. A negative result on bolt-on matrix latents (Larson, 2026), where a flatten-then-project readout makes rank invisible to the gradient, is this result's negative counterpart.

Limitations. All results are on sub-1M-parameter synthetic testbeds; nothing here is a claim about pretrained language models. Over-recruitment beyond two spare dimensions is not expressible in this state window; razor cells are single-seed except S_3 (four seeds; necessity zeros exact wherever seeds exist); two groups at the shortest pinned budget show soft convergence (disclosed; the razor reading is anchor-relative and unaffected).

G	$\lvert G \rvert$	$d_{\min}$	solvable	realization of ρ_G
S_3	6	2	yes	standard (dihedral)
S_4	24	3	yes	cube rotations
A_5	60	3	no	icosahedral rotations
S_5	120	4	no	standard (zero-sum)
A_6	360	5	no	standard (zero-sum)

Table 1: The five task groups. $d_{\min}$ is the minimal faithful real representation dimension; each reference representation is orthogonal by construction and verified faithful (injective on G) at build time. S_4/A_5 form the marquee pair: matched $d_{\min}$, opposite solvability.

G	$d_{\min}$	n	restricted eff. rank	band	in band
S_3	2	3	1.877 ± 0.060	$[1.4, 2.6]$	3/3
S_4	3	5	2.852 ± 0.054	$[2.1, 3.9]$	5/5
A_5	3	5	2.832 ± 0.062	$[2.1, 3.9]$	5/5
S_5	4	3	3.591 ± 0.069	$[2.8, 5.2]$	3/3
A_6	5	3	4.736 ± 0.023	$[3.5, 6.5]$	3/3

Table 2: Restricted effective rank of the unconstrained trained state (mean $\pm$ sd over seeds) against each group's $d_{\min}$, with the pre-registered $[0.7, 1.3] \cdot d_{\min}$ band. Every seed of every group is in band; Spearman $\rho = 0.9747$ is the tie-capped maximum; the dimension-matched pair S_4/A_5 differs by 0.019 rank-units.

Appendix A. The Five Groups and Reference Representations

Appendix B. Per-Seed Observational Rank

The instrument's centering step is load-bearing: an uncentered lens scores a flawless synthetic model 0.705 where the centered production lens scores 0.9996 on identical data. The disclosed length-robustness split referenced in Section 3 scores only words of length $L \geq 2$ (the $L = 1$ read is attention-degenerate and was demoted with a mechanism note in the run records): it moves per-group mean recovery cosine by at most 0.013 (per-seed at most 0.041), with no group-selective divergence.

Appendix C. The Ambient-Identity Tax in Detail

The first sweep's force-rank target was the eye-padded $\rho_G(\cdot) \oplus I_2$, whose rank is $d_{\text{state}} = d_{\min} + 2$ for every group. Because the identity block is constant across words, it is the cheapest reliable loss reduction available, and a rank-capped arm buys it before the group representation; the residual budget for ρ_G is $k - 2$, below $d_{\min}$ at every grid point, so no capped cell in the original sweep ever tested the law's confirm direction. Three raw-artifact signatures established this: (i) force-rank cells' direct cosine matched the rank-k optimum $\sqrt{k/d_{\text{state}}}$ with mean absolute deviation 0.028 across all 39 cells (maximum 0.166; the two

S_5 below-$d_{\min}$ cells are the only outliers, an additional disclosed optimization shortfall), so the arms trained to their rank-constrained ceiling rather than under-training; (ii) centered restricted rank of capped arms landed near $k-2$, the residual budget; (iii) in the repaired wave, an eye-padded corroboration arm reproduces the failure on demand at raw $k = d_{\min}+1$ (effective budget $d_{\min}-1$) while the tax-paid point recovers. The repaired wave's 30 cells were configuration-verified one-by-one against a manifest re-derived independently from the design record (steps, padding mode, and force-rank value per cell; zero skipped steps).

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